

2 Hardware device files

In Linux all the hardware devices are treated as files. They allow transparent communication between user space applications and hardware devices. A device file is an interface for a device driver that appears as if it were an ordinary file. Device files provide simple interfaces to peripheral devices such as printers in addition to allowing access to specific resources on devices such as hard disk partitions. In addition to providing interfaces to hardware devices, device files provide access to system resources that have no actual connection with any actual device such as random number generators or data sinks. Device files allow users or software to interact with device drivers using system calls or standard input/output. This simplifies several tasks.

There are two types of device files, character devices and block devices:

Character devices communicate to devices through which the system transmits data one character at a time. Examples of such devices are serial modems, virtual terminals, mice and keyboards. Such devices usually do not support random access to data.

Block devices relate to devices through which data is moved in the form of blocks (one block can range from 512 bytes to 32 kilobytes). Examples of such devices are hard disks, CD or DVD ROMs and memory regions.

All the device files are located in the `/dev` directory. You can find out the type of a particular device file by using `ls -l` or `file`. If you use `ls -l` the first letter of the permission string shows the device file's type. If it is `c`, then it means that it is a character special file and if it is `b` it means that it is a block special file. You can also use `cat /proc/devices` to get a list of all character and block files.

Example:

```
$ ls -l /dev/tty1
crw----- 1 jck tty 4, 1 Jun 15 21:08 /dev/tty1
$ file /dev/tty1
/dev/tty1: character special

$ ls -l /dev/sda1
brw-rw---- 1 root disk 8, 1 Jun 15 18:08 /dev/sda1
$ file /dev/sda1
/dev/sda1: block special
```

2.1 Naming conventions

These prefixes are commonly used in Linux systems to identify device drivers in the `/dev` hierarchy:

1. fb: frame buffer (video output device)